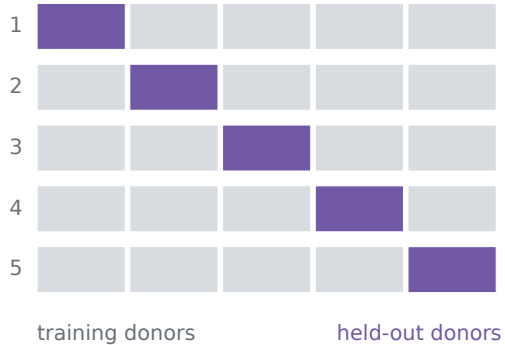


A DATA AND BIOLOGICAL HOLDOUT

Atlas	Cells	Donors	Labels
TNBC	427,813	101	7
PBMC	462,034	199	13
Brain	75,580	12	10

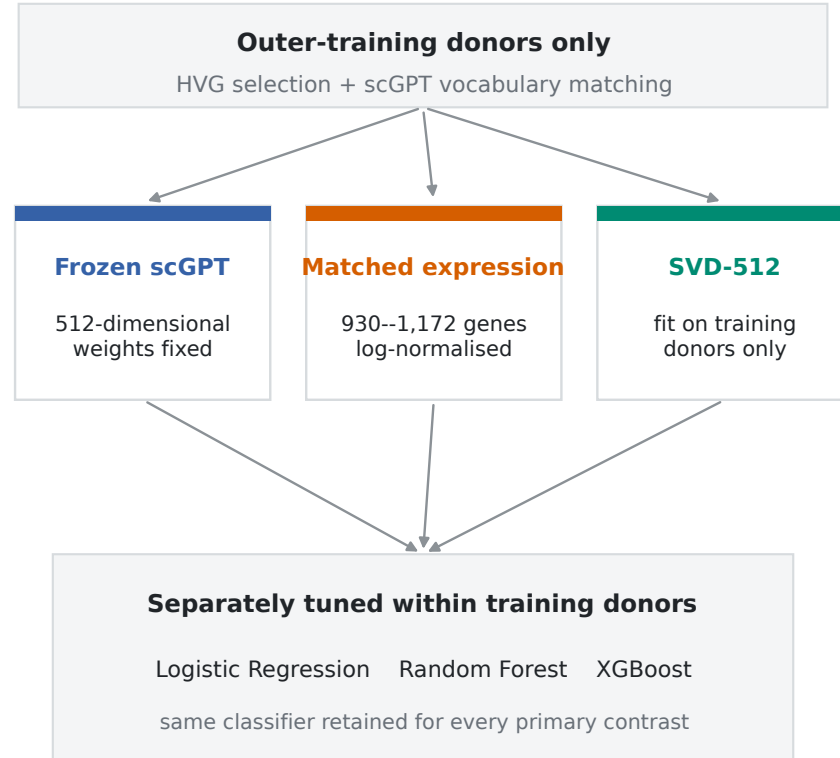
965,427 analysed cells | 312 donors

Five outer folds per atlas



Each donor is tested once; train/test donors never overlap.

B REPRESENTATION-MATCHED LEARNING



Locked outer-test predictions generated once

C PRESPECIFIED EVALUATION

PRIMARY

Macro F1
fixed-classifier contrasts

UNCERTAINTY

paired donor bootstrap
5,000 resamples

PROBABILITIES

ECE, Brier, NLL
temperature scaling

BIOLOGY

class F1 + confusions
strict label scarcity

SEPARATE STRESS TEST

pig pancreas; random cell split